

SEMESTER-V

UMT403: STRESS ANALYSIS AND MECHANICAL DESIGN				
	L	T	P	Cr
	3	1	2*	4.0
Course Objective: This subject aims to develop an understanding of the stresses and strains that develop in solid materials when they are subjected to different types of loading and to develop an understanding of the conditions at failure of such materials. Further, it provides students with the ability to apply design procedure for machine elements with specific design tools representing empirical, semi-empirical and analytical approaches.				
Syllabus Axial Stresses: Deformation of axial loaded uniform/stepped bars, axial stresses in bars, stress-strain relationships; Young's modulus of elasticity, shear modulus and Poisson's ratio; two-dimensional elasticity; isotropic and homogeneous materials; ductile and brittle materials; buckling of columns Bending and Shear Stresses: Bending stress calculation for beams of simple and built up sections; shear stress formula for beams, shear stress distribution in beams; torsion of shafts. Stress Transformation and Principal Stresses: Stresses on an arbitrary inclined plane, Mohr's stress circle, and principal stresses, principal strains, strain gauges and strain rosettes. Design Processes: Introduction, standards and preferred numbers, selection of materials, theories of failure, design against static loads, stress-concentration, and design against dynamics loads, factor of safety and its selection. Design of Shaft: Shafts subject to combined static loading. Design of Gears: Introduction to types, force analysis and application; Gear tooth failure, beam strength and wear strength of gear tooth, materials and manufacture. Brake: Introduction to types, construction, application and force analysis, band brakes, block brakes, expanding shoe brakes.				
Laboratory Work : The laboratory work involves 1. Determination of tensile strength and Young's modulus 2. Determination of shear modulus (non-destructive torsion test) 3. Determination of shear strength and shear modulus (destructive torsion test) 4. Determination of impact strength				
Course Learning Outcomes (CLOs) The students will be able to: 1. evaluate various kinds of stresses (axial, bending, torsional and shearing) in various structural elements due to different type of external loads. 2. determine stresses on an arbitrary inclined planes and locate principal planes and stresses. 3. conduct a failure analysis for the design/sizing of mechanical components. 4. determine suitable module and specifications of gears from strength and wear considerations				
Text Books 1. Popov, E.P. and Balan, T.A., Engineering Mechanics of Solids, Prentice Hall of India (2012). 2. Rattan, S.S., Strength of Materials, McGraw Hill Education, Chennai 3rd edition (2017)				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

3. Shigley, J., Mechanical Engineering Design, McGraw Hill, New York (2014), 10th edition
4. Bhandari, V. B., Design of Machine Elements, Tata McGraw Hill, New Delhi (2010), 3rd Edition.
5. Norton, R.L., Machine Design: An Integrated Approach, Pearson Education, New Delhi (2009), 3rd edition

Reference Books

1. Hibbeler, R.C., Mechanics of Materials, Pearson, 11th impression (2018).
2. Shames, I. H. and Pitarresi, J. M., Solid Mechanics, Prentice Hall of India (1996).
3. Juvinall, R. C. and Marshek, K. M., Fundamental of Machine Component Design, John Wiley & Sons, New York (2011), 5th edition
4. Spotts, M. F. and Shoup, T. E., Design of Machine Elements, Pearson Education, New Delhi (2003)
5. Design Data: Data Book Of Engineers By PSG College - Kalaikathir Achchagam – Coimbatore (2012)
6. Mahadevan, K. Balaveera Reddy K., Design Data Handbook for Mechanical Engineers in SI and Metric Units, CBS Publishers & Distributors (2018)

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1	MST (2 hrs.)	40
2	EST	30
3	Lab Evaluation	20
4	Sessional (tutorial/quiz, etc.)	10

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.